

Preserving School Air as Admissible Evidence

An independent technical governance reference on the Chiang Mai classroom air-quality comparison, 2025–2026

STATUS	DOCUMENT CONTROL
PRE-EXAMINATION · HOLD	Version 2.0 8 October 2026
Baseline not frozen	Independent educational reference; no certification or institutional endorsement

The question that changes the conversation

Air-quality measurement can support better decisions. But a measured improvement, a comparative study, and permission to implement a consequential intervention are three different things. TA14 examines the boundary between them.

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Institutional independence. AirGradient, the school, its operators, and researchers have not commissioned, reviewed, approved, or endorsed this reference. TA14 has not received or audited the underlying dataset. No engineering approval, product certification, causal finding, or health-outcome assurance is issued.

01 | What the published study actually reports

AirGradient describes an observational comparison at one Chiang Mai school, using weekday observations from 08:00 to 15:00 during the available March–April periods in 2025 and 2026. The comparison groups are spaces with air purifiers and spaces with professionally installed positive-pressure systems (PPS).

Mean PM2.5 (µg/m³)	2025	2026
Outdoor reference	58.7	87.8
Air purifier spaces	24.2	17.4

Mean PM2.5 (µg/m³)	2025	2026
Included PPS spaces	3.4	5.0

The included PPS group averaged approximately 86% lower PM2.5 than the purifier group in 2025 and 71% lower in 2026. These are observed group differences, not a randomized causal comparison or a guarantee for other buildings.

02 | What must remain attached to those numbers

Boundary	Published qualification
Coverage	33 recorded dates in 2025; 44 in 2026. Three outdoor monitoring locations.
Compared spaces	Nine purifier spaces each year; 40 included PPS spaces in 2025 and 43 in 2026.
Exclusions	Eight malfunctioning or non-maintained PPS installations were excluded.
Study design	Rooms were not randomly assigned; differences in room characteristics, system sizing, operation, and maintenance may matter.
Threshold interpretation	EPA “Good” reference bands were applied to opening-hours daily averages, not official 24-hour AQI. The WHO annual guideline cannot establish annual compliance from smoke-season means.

The excluded installations are not disposable history. They are part of the operational record needed to assess reliability and transferability. Their exclusion from a defined performance analysis may be justified, but the exclusions and rationale must remain visible.

03 | From observation to admissible consequence

TA14 anchor	School-air evidence question
REALITY	What actually occurred in occupied rooms, with smoke, doors, ventilation, and maintenance?
RECORD	Which monitors, locations, times, raw readings, corrections, and quality flags?
CONTINUITY	Can transformations, missing periods, replacements, and exclusions be reconstructed?
ADMISSIBILITY	Is the evidence fit for this particular claim and decision?
BINDING	Which specific room, asset, owner, period, and intervention does the evidence concern?
COMMIT	Who has established authority to approve the bounded decision?
EXECUTION	Was the intervention performed under applicable local authority and controls?
OUTCOME	What changed afterward, including adverse and unexpected results?

Observed is not inferred. Inferred is not authorized. An admissible input does not create an admissible output. Capability $\neq$ Authority.

04 | The minimum evidence package

Measurement provenance. Device IDs, site/room mapping, timestamps and time zone, calibration or correction methods and versions, sampling cadence, outdoor placement, missingness, and quality flags.

Physical and operational context. Ventilation and filtration specifications, fan and airflow performance, room enclosure and pressure behavior, control sequences, occupancy, door practices, and operating schedules.

Exception continuity. The eight excluded PPS installations: identification, exclusion rationale, failure mode, maintenance history, dates, and sensitivity of conclusions to inclusion or exclusion.

Analytical reproducibility. Frozen dataset, data dictionary, daily aggregation rules, group denominators, regression inputs, uncertainty, and separation of direct observation from interpretation.

Authority and consequence. Named decision-maker, local engineering and safety requirements, owner/operator approval, proposed action, stop/rollback conditions, and verification responsibility.

05 | One bounded demonstration, not a blanket endorsement

Proposed proposition: Can the published classroom PM2.5 comparison be traced through a frozen evidence record sufficiently to inform one defined school-level decision, with its exclusions and limitations intact?

Step	Boundary and output
INVITE	Invite corrections to this characterization; participation is voluntary.
ESTABLISH	Identify a requester with standing, registered governance, a narrow proposition, and a frozen baseline.
EXAMINE	Test provenance, comparability, exclusions, uncertainty, and claim-to-decision boundaries.
DISPOSE	Record ALLOW / HOLD / DENY / ESCALATE only within the examination scope. No outcome is promised.
PUBLISH	With permission, separate source observations, TA14 findings, unresolved issues, and any locally authorized actions.

Illustrative decision test

A different school proposes installing PPS because of the Chiang Mai results. Its local room enclosure, smoke exposure, filtration, energy, acoustics, maintenance capacity, budget, codes, and accountable engineering review are not established by the published comparison. An illustrative TA14 response could be HOLD pending local evidence and approvals. This is not a finding against AirGradient or the Chiang Mai school.

06 | Questions worth discussing with AirGradient

- Can the published group means and exclusion decisions be reproduced from a frozen, appropriately shared dataset?
- Which measurements were directly observed, corrected, inferred, or unavailable?
- What can the eight excluded PPS records tell us about operational reliability and maintenance?
- What decisions are being made from these monitoring results, and who is authorized to make them?
- What evidence would another school need before acting on these findings?

An invitation, not a demand

TA14 proposes a voluntary technical conversation to identify one bounded evidence-to-consequence question. A formal examination would require registered governance, a frozen baseline, defined participation, and established standing. No partnership, endorsement, or access to nonpublic records is presumed.

Greggory Don Butler | Founder, TA14 Authority Governance Institution
www.ta14exchange.com

Source and attribution

Achim Haug, “Air Purifiers vs Positive Pressure Systems: Field Performance Across Two Smoke Seasons in Classrooms,” AirGradient, 20 September 2026. [Read the source article](#). All reported measurements and methodological descriptions above are attributed to this article. TA14 examination concepts and proposed tests are independent framing.